(KTXFI1EBNF) Physics I. Lecture

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International Women's Day

Happy International Women's Day!

Celebrating the strength, beauty, and wisdom of women everywhere.

Current Updates and Issues

- Materials for the previous lecture and practice session have been uploaded to Moodle, YouTube, and GitHub.
- Short videos have been added to Moodle to help clarify the underlying physics.
- Homework No. 1 has been released (Deadline: March 22, 2026, midnight). I also provided some supplementary material due to the limited practice time.
- ~~Grave~~ood news: The job offer from ISU was withdrawn without explanation; therefore, I will be staying here until the summer.
- Proposed midterm exam date: **May 11, 2026**. The retake will be on **May 18, 2026**.

Where are we? I

1. **Mechanics I.:** Newton's laws, equation of motion (review). Momentum. General concept of mechanical work, work theorem. Forms of mechanical energy.
2. **Mechanics II.:** Weight force, gravitational force, gravitational field. Work of the gravitational force. Conservative force field. Dissipative force field. Mechanical power.
3. **Mechanics of Rigid Bodies I.** Concept of a rigid body. Forces acting on a rigid body. Motion of a rigid body. Concept of the centre of mass. Location of the centre of mass. Torque. Angular momentum. Energy of a rotating rigid body, angular momentum of a rotating rigid body, Steiner's theorem.
4. **Mechanics of Rigid Bodies II.** Principal theorems of rigid body motion; conservation laws. Collisions.
5. **Oscillations I.** Dynamic condition of harmonic oscillatory motion. Differential equation of harmonic undamped oscillatory motion.

Where are we? II

6. **Oscillations II.** Superposition of oscillations. Decomposition of oscillations into components. Damped oscillations. Forced oscillations, resonance. Coupled oscillations.
7. **Wave phenomena:** Reflection of waves, refraction of waves, diffraction of waves. Huygens–Fresnel principle. Interference of waves. Polarization of waves.
8. **Elements of acoustics:** Basics of acoustics. Fundamental concepts. Doppler effect. Sound intensity, loudness according to Weber–Fechner's law.
9. **Thermodynamics I:** *Temperature scales, thermal expansion, thermometers. State variables, ideal gas, state changes, ideal gas equation of state, Boyle's law, Gay-Lussac's laws, universal (combined) gas law, heat, specific heat, molar heat, heat capacity.*
10. **Thermodynamics II:** Expansion work, internal energy, the first law of thermodynamics, forms of the first law in different state changes.

Where are we? III

11. **Thermodynamics III:** Cyclic processes, Carnot cycle, Carnot's theorem, reduced heat, entropy, the second law of thermodynamics.
12. **Thermodynamics IV:** Kinetic theory of gases, molecular thermodynamics. Distributions, distribution functions. Elementary statistics.
13. **Electrostatics I.:** Electric charge: Fundamental electrical phenomena, electric charge and matter, Coulomb's law (force between two point charges, unit of electric charge, permittivity of vacuum).
14. **Electrostatics II.:** The electric field: The electrostatic field (electric field strength, electric field lines). Determination of electric field strength. Gauss's law (flux of the electric field). Electric field of a point charge.

Where are we? IV

15. **Electrostatics III.:** Electrostatic potential: Work of the electric field (work done in an electric field). Electric potential (electrostatic potential). Electric potential difference, voltage. Electric potential of charge systems at small and large distances. Electrostatic energy.
16. **Elements of physical or wave optics.** Branches of optics. Light interference, light diffraction, diffraction through a slit, optical grating, light polarization. Light as an electromagnetic wave: Poynting vector. Optical instruments.

Thermodynamics – Temperature scale I

- Thermodynamics
  - Phenomenological thermodynamics – macroscopic description
  - Kinetic theory of gases – microscopic description

- CELSIUS temperature scale
  - Anders Celsius (1701 - 1744)
  - **Originally:**
    - $100^\circ C$: the freezing point of water in normal atmospheric pressure
    - $0^\circ C$: the boiling point of water in normal atmospheric pressure
  - **Later and today:**
    - $100^\circ C$: the boiling point of water at normal atmospheric pressure
    - $0^\circ C$: the freezing point of water in normal atmospheric pressure

Thermodynamics – Temperature scale II

- FAHRENHEIT temperature scale
  - Daniel Fahrenheit (1686 - 1736)
  - The first person who used mercury thermometer
  - **Key points of the scale:**
    - $32^\circ F$: the freezing point of water at normal atmospheric pressure ($0^\circ C$)
    - $212^\circ F$: the boiling point of water at normal atmospheric pressure
  - **Conversion rule from Celsius to Fahrenheit:**

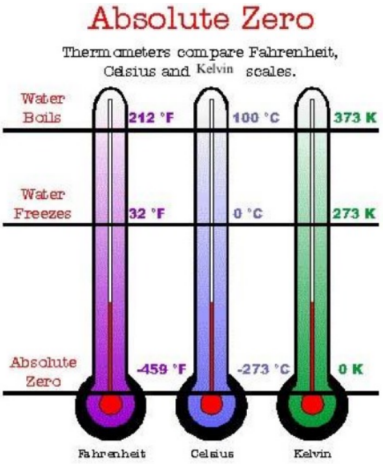
$$
\begin{aligned}
t(^\circ\mathrm{F}) &= \frac{180}{100} t(^\circ\mathrm{C}) + 32 \\
t(^\circ\mathrm{F}) &= \frac{9}{5} t(^\circ\mathrm{C}) + 32
\end{aligned}
$$

Thermodynamics – Temperature scale III

- KELVIN temperature scale – absolute temperature scale
  - Lord Kelvin (1824 - 1907)
  - The size of the division is the same as on the Celsius scale: $\Delta T = \Delta t$
  - **Conversion rule from Celsius to Kelvin:**

$$
\begin{aligned}
T(\mathrm{K}) &= t(^\circ\mathrm{C}) + 273,15 \\
T(\mathrm{K}) &\approx t(^\circ\mathrm{C}) + 273
\end{aligned}
$$

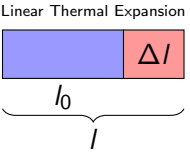
Thermodynamics – Temperature scale IV



Thermodynamics – Thermal expansion I

- Linear thermal expansion

$$
\begin{aligned}
\Delta l &\sim l_0 \\
\Delta l &\sim \Delta T \\
\Delta l &= \alpha \cdot l_0 \cdot \Delta T \\
l &= l_0 \cdot (1 + \alpha \cdot \Delta T)
\end{aligned}
$$



α: linear thermal expansion coefficient, $[\alpha] = 1/\mathrm{K}$

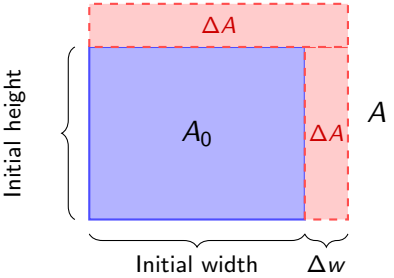
Thermodynamics – Thermal expansion II

- Surface thermal expansion

$$
\begin{aligned}
\Delta A &\sim A_0 \\
\Delta A &\sim \Delta T \\
\Delta A &\sim A_0 \cdot \Delta T \\
\Delta A &= 2 \cdot \alpha \cdot A_0 \cdot \Delta T \\
\Delta A &= A - A_0 \\
A &= A_0 \cdot (1 + 2 \cdot \alpha \cdot \Delta T)
\end{aligned}
$$

α: linear thermal expansion coefficient, $[\alpha] = 1/\mathrm{K}$

Surface Thermal Expansion



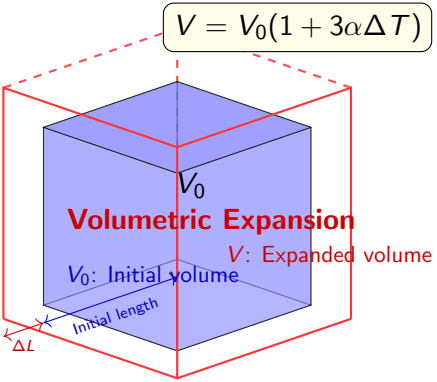
$$\boxed{A = A_0(1 + 2\alpha\Delta T)}$$

Thermodynamics – Thermal expansion III

- Volumetric thermal expansion

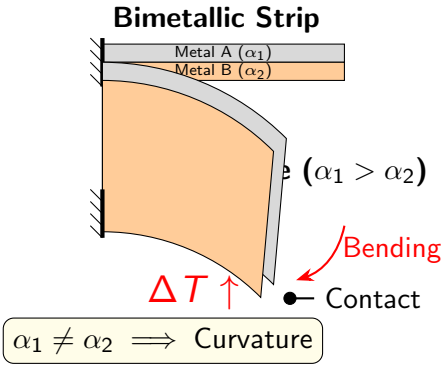
$$
\begin{aligned}
\Delta V &\sim V_0 \\
\Delta V &\sim \Delta T \\
\Delta V &\sim V_0 \cdot \Delta T \\
\Delta V &= 3\alpha \cdot V_0 \cdot \Delta T \\
\Delta V &= V - V_0 \\
V &= V_0 \cdot (1 + 3\alpha \cdot \Delta T) = V_0 \cdot (1 + \beta \cdot \Delta T)
\end{aligned}
$$

β: volumetric thermal expansion coefficient, $\beta = 3\alpha$



Thermodynamics – Thermometer I

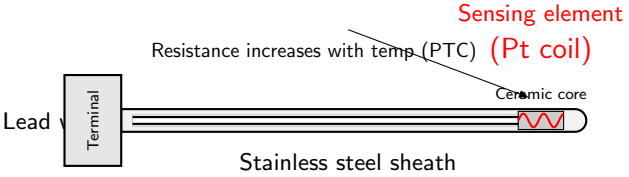
- Bimetal
  - Based on thermal expansion
  - The bending effect caused by the difference in thermal expansion of two materials with different material properties is used to measure temperature.
  - Bimetellic thermometers use the bending effect caused by the difference in thermal expansion of the two metals to measure temperature.
  - It is used to measure the effects of temperature change. The bimetal strip bends in response to changes in temperature
  - Two different kinds of thermal expansions
  - Used for fire detection



Thermodynamics – Thermometer II

- Resistance thermometer

$$R(T) = R_0(1 + \alpha \Delta T)$$



- Based on thermal expansion
- Solid-state thermometers
- The resistance of a metal wire changes with temperature change
- Platinum resistance thermometer: the changes of the resistance of the platinum is used for temperature measurement
- Wide range of linearity

Thermodynamics – Thermometer III

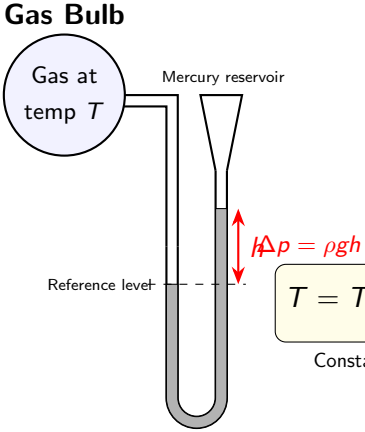
- Two types of semiconductor solid-state thermometer:
  - NTC (Negative Temperature Coefficient): as the temperature increases, the resistance of an NTC thermistor decreases. This negative correlation between resistance and temperature gives it its name.
  - PTC (Positive Temperature Coefficient): in contrast to NTC thermistors, the resistance of a PTC thermistor increases as the temperature rises. The positive correlation gives it its name.

Thermodynamics – Thermometer IV

- Liquid thermometers: Mercury or alcohol thermometer
  - Based on thermal expansion
  - Liquid thermometers: Mercury or alcohol thermometer
  - The liquid thermal expansion is used for temperature measurement
  - The thermometer has a uniformly spaced scale.
  - The alcohol is placed in an expansion vessel, which is connected to a capillary tube. The higher the temperature, the higher the alcohol rises in the capillary tube.
  - For example: traditional room thermometer

Thermodynamics – Thermometer V

- Gas thermometers, Constant-pressure gas thermometer:
  - Based on thermal expansion
  - The has thermal expansion
  - The change in volume of the confined gas is read.
  - The mercury droplet rises in the thermometer as the temperature increases.



$$T = T_{tr} \cdot \frac{p}{p_{tr}}$$

Constant volume: $V = \text{const.}$

Mercury Manometer

Thermodynamics – Thermometer VI

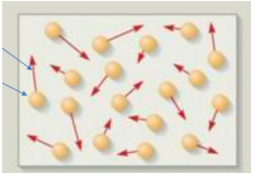
- Special thermometers
- **Laser based thermometer:**
  - Every object emits thermal radiation, the wavelength of which depends on the temperature of the objective.
  - A major advantage is that there is no direct contact with the object being measured.
- **Colour thermometer:** e.g., used in metallurgy

Phenomenological Thermodynamics I

- State indicators
  - **Volume (V):** $[V] = m^3$
  - **Pressure (p):** $[p) = Pa$
  - **Temperature (T):** $[T] = K$
  - **Amount of gas (n):** $= mol$

State indicators: p, V, T, n

- **The ideal gas is a model:**
  - Tiny, point-like particles (small spheres).
  - Constant, random motion.
  - Perfectly elastic collisions with each other and the walls.
  - The volume of particles and long-range interactions are neglected.
  - No attraction between gas particles.



Phenomenological Thermodynamics II

- The ideal gas equation of state

$$p \cdot V = n \cdot R \cdot T$$

  where p: pressure, V: volume, n: amount of substance (material), R: universal constant, T: temperature.
- **Universal gas constant:** $R = 8.314 \frac{J}{mol \cdot K}$
- **Alternative forms:**

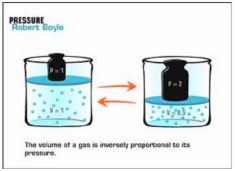
$$p \cdot V = \frac{m}{M} \cdot R \cdot T$$
$$p \cdot V = m \cdot R_i \cdot T$$

  Where M is molar mass, m is mass, and $R_i = \frac{R}{M}$ is the individual gas constant.

Phenomenological Thermodynamics III

- Special gas laws
  - **Isothermal (T = const.):** Boyle – Mariotte Law: When the temperature of a given mass of confined gas is constant, the product of its pressure and volume is constant. During a state change at constant temperature (isotherm), the pressure and volume of an ideal gas are inversely proportional to each other.

$$p_1 \cdot V_1 \quad = \quad p_2 \cdot V_2$$



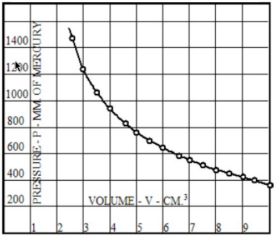
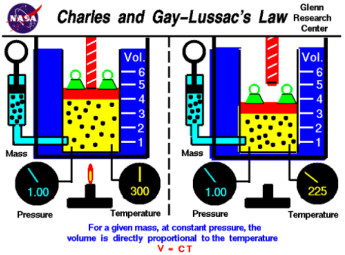


Fig. 1. Curve showing the variation of the volume of a gas at constant temperature as a function of the pressure.

Phenomenological Thermodynamics IV

- **Isobaric (p = const.):** Gay-Lussac I. Law: During a state change at constant pressure (isobaric), the volume and temperature of an ideal gas are directly proportional to each other; that is, the ratio of the volume to the temperature of the ideal gas is constant.

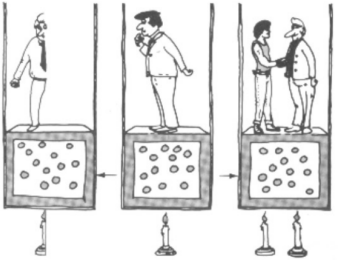
$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$



Phenomenological Thermodynamics V

- **Isochoric (V = const.):** Gay-Lussac II. Law: During a state at constant volume (isochoric), the pressure and temperature of an ideal gas are directly proportional to each other; that is, the ratio of the pressure to the temperature of the ideal gas is constant

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

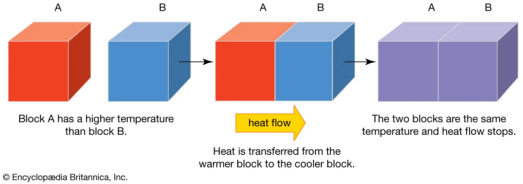


- UNIVERSAL GAS LAW: If the amount of substance n is constant:

$$\frac{p_1 \cdot V_1}{T_1} = \frac{p_2 \cdot V_2}{T_2} \tag{1}$$

Heat, Specific Heat, Heat Capacity I

- Joseph Black (1728-1799) Experiments and observations on heat processes.
  - Heating a piece of copper and water on the same heating source (with the same surface area, same heat transfer), they reach different temperature in the same amount of time.
  - Different materials behave differently under the same amount of heat.
  - Heat has quantity.
- Materials in contact with each other: **Heat flows, transfers from one material to another**



Heat, Specific Heat, Heat Capacity II

- **Specific heat (c):** The heat required to raise the temperature of 1 kg of substance by $1°C$.

$$\begin{aligned} Q &= c \cdot m \cdot \Delta T \\ c &= \frac{Q}{m \cdot \Delta T} \\ c &= \frac{J}{kg \cdot °C} = \frac{J}{kg \cdot K} \end{aligned}$$

- Specific heat of ideal gases
  Relationship: $c_p - c_V = R$

Type of gas	c_V	c_p
One-atomic gases (He, Ne, Ar, Kr, Xe, ...) – noble gases	$\frac{3}{2}R$	$\frac{5}{2}R$
Two-atomic gases (H_2, O_2, N_2, ...)	$\frac{5}{2}R$	$\frac{7}{2}R$
More-atomic gases (CH_4, ...)	$\frac{7}{2}R$	$\frac{9}{2}R$

Heat, Specific Heat, Heat Capacity III

- **Heat Capacity:** The numerical value of the heat capacity indicates how much heat is required to raise the temperature of the body by 1 C degree. Symbol: K, dimension: J/K, J/C°.

$$K = \frac{Q}{\Delta T} = c \cdot m$$

$$K = \frac{J}{K} = \frac{J}{^{\circ}C}$$

The End

Thank you for your attention!

Acknowledgements Thank you for the course materials for Szilárd ZSÓKA and Zoltán VARGA.